

AI-Powered Proactive Traffic Management and Intelligent Route Recommendation System

Knowledge and Semantic Computing
Final Year Project

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Problem Statement

Current Problems

- Increasing traffic congestion
- Inefficient traffic signal control
- Delayed emergency vehicle movement
- Lack of real-time traffic prediction
- No intelligent alternate route recommendation

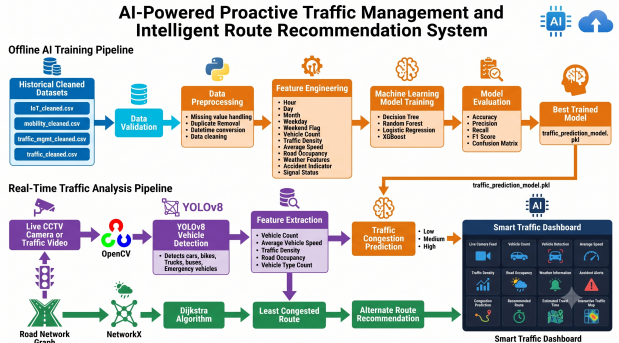
Existing Traffic Management Techniques

- **Fixed-Time Traffic Signals**
 - ▶ Operate using fixed timers.
- **Inductive Loop Sensors**
 - ▶ Detect vehicles using road-embedded sensors.
- **Computer Vision**
 - ▶ Uses CCTV cameras to detect vehicles and traffic density.
- **IoT-Based Monitoring**
 - ▶ Collects live traffic data from smart sensors.
- **Adaptive AI Traffic Control**
 - ▶ Dynamically adjusts signal timings based on real-time traffic.

- Existing systems mainly react after congestion occurs.
- Most systems do not predict future traffic conditions.
- Route recommendations are often based on current traffic only.
- There is limited integration of AI prediction with intelligent route planning.
- Our system combines Computer Vision, Machine Learning, and Graph Algorithms to provide proactive traffic management.

Proposed Solution

- AI-based traffic prediction
- Live vehicle detection
- Real-time Congestion Prediction
- Smart route recommendation
- Interactive dashboard



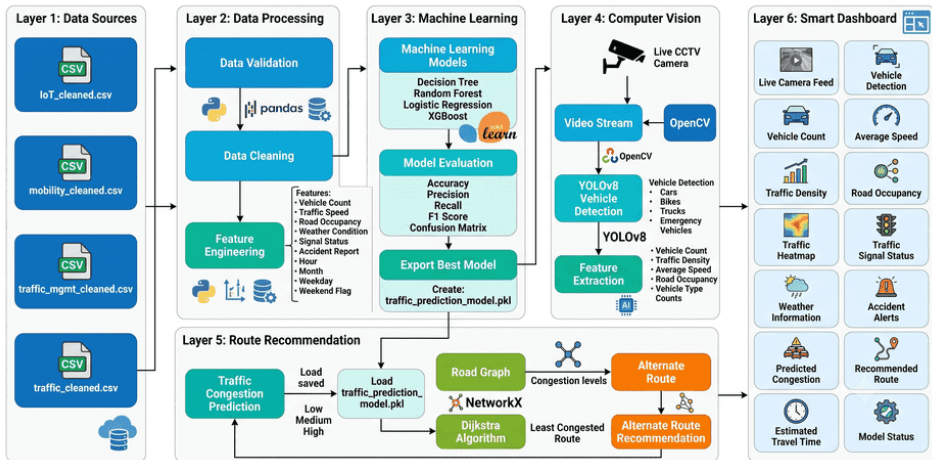
Objectives

- ① Detect vehicles using Computer Vision (YOLOv8)
- ② Predict traffic congestion using Machine Learning
- ③ Recommend optimal routes using NetworkX and Dijkstra Algorithm
- ④ Improve real-time traffic management
- ⑤ Develop Smart City dashboard

Dataset	Purpose	Status
IoT_cleaned.csv	Live Traffic Features	Preprocessed
mobility_cleaned.csv	Machine Learning Training	Preprocessed
traffic_mgmt_cleaned.csv	Feature Analysis	Preprocessed
traffic_cleaned.csv	Testing / Validation	Preprocessed

System Architecture

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Technology Stack

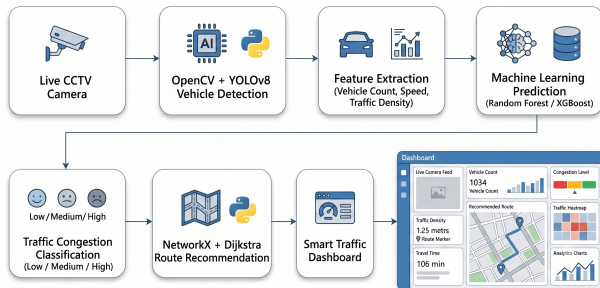
Category	Technology
Programming	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, XGBoost
Computer Vision	OpenCV, YOLOv8
Routing	NetworkX, Dijkstra Algorithm
Dashboard	Streamlit
Visualization	Plotly
Model Storage	Pickle (.pkl)
Machine Learning Models	Random Forest

Expected Output

System Features

- Live Vehicle Detection
- Traffic Density Analysis
- Congestion Prediction
- Smart Route Recommendation
- Interactive Dashboard

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Current Progress

Module	Status
Dataset Collection	Completed
Data Cleaning	Completed
Feature Engineering	Completed
Machine Learning Training	In Progress
YOLOv8 Integration	Planned
Route Recommendation	Planned
Dashboard Development	Planned

- Train ML models
- Deploy YOLOv8
- Integrate NetworkX
- Build Streamlit Dashboard
- Validate the system using real-time traffic scenarios and evaluate prediction accuracy.

References

- Smart Mobility Dataset (Kaggle)
- IoT Traffic Flow Dataset (Kaggle)
- Ultralytics YOLOv8 Documentation
- OpenCV Documentation
- Scikit-learn Documentation
- NetworkX Documentation

Thank You